

MEASURING SOCIO-ECONOMIC BEHAVIOR IN INTERNET GAMES

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We analyze high-frequency log-files of a massive multiplayer online game currently played by thousands of users. This allows us to assess socio-economical dynamics over a time-span over the past three years. We can demonstrate that a series of stylized facts present in real-world social systems can be one-to-one related to the social and economic behavior of the players in the game. In particular, we analyze the evolution of underlying growing social networks such as constituted by cooperators or defectors and private message communication networks, and measure their characteristic properties. Since the game focuses heavily on (virtual) trading between the players, we derive a series of economical timeseries and compare them to real-world data. The similarities found in the economic context give hope to establish a sound "laboratory" for social and economical behavior.

Keywords

Network analysis, time series analysis, social interaction